

Suggestions for Jiya Research

Your present architecture is

CPCB stations + ERA5 BLH + wind + inversion + FIRMS fires → dynamic graph → GAT/STGNN → PM_{2.5} forecast

But

Dynamic graph + wind + GNN + BLH is no longer sufficiently novel by itself.

Recent work has already introduced dynamic geographical graphs using wind fields for PM_{2.5} forecasting, and other studies have used directed graphs to model regional transport.

<https://www.sciencedirect.com/science/article/abs/pii/S1364815225000350>

There is even very recent work proposing a transport-informed GNN based on advection diffusion reaction operators for PM_{2.5} forecasting.

<https://www.nature.com/articles/s44407-026-00052-x>

And dynamic graph construction for air-quality forecasting is already appearing in patent literature.

<https://patents.google.com/patent/CN120724852B>

It is better to upgrade the research into

A physics-constrained, source-aware, multi-scale dynamic transport graph network for PM_{2.5} forecasting

Do not make the invention about a GNN. Make the invention about a new pollution-transport representation and forecasting mechanism, with the GNN as the implementation.

Your current research is

wind + BLH + inversion → dynamically change graph connections.

That is useful, but we should go one level deeper.

Pollution movement should be represented as

Emission/source → atmospheric transport → stability-controlled dispersion → receptor accumulation → PM_{2.5} forecast

So the model becomes a source-to-receptor transport system, rather than simply a station-to-station correlation system.

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Your research already moves in this direction by proposing Punjab/Haryana/Rajasthan as external source nodes. That is the right idea. But we should make it mathematically much stronger.

Your current research proposes

$BLH < 100 \text{ m} \rightarrow$ suppress long-distance edges and increase self-loop.

I would not use a hard 100-m rule as the central mechanism.

Why?

Because a reviewer can easily ask:

Why exactly 100 m? and Is pollution transport really discontinuous at 100 m?

The answer is no.

Instead use a continuous Atmospheric Stability Index.

For example:

$$S_t = f(B, L, H_t, T_{inversion,t}, RH_t, WS_t, SR_t)$$

where:

- BLH = boundary-layer height
- $T_{inversion}$ = inversion strength
- RH = relative humidity
- WS = wind speed
- SR = solar radiation

Then:

$$0 \leq S_t \leq 1$$

where:

- $0 \rightarrow$ strong atmospheric mixing
- $1 \rightarrow$ strong atmospheric trapping

Now the graph is continuously modified:

$$A_{ij}^t = A_{ij}^{wind,t} \cdot G(S_t) \cdot T_{ij}^t$$

This is much stronger than:

"If $BLH < 100 \text{ m}$, turn edges off."

Add transport travel time

This is where I think your research can become significantly stronger.

At present, the concept mostly says:

Wind direction $\rightarrow$ connect source to station.

But pollution does not teleport from Punjab to Delhi.

It has a travel time.

Therefore introduce:

Dynamic Transport Lag

For source and receptor i :

$$\tau_{s,i}(t) = \frac{d_{s,i}}{U_{eff}(t)}$$

where:

- $d_{s,i}$ = effective transport distance

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- U_{eff} = effective wind transport velocity

Then source influence becomes:

$$E_s(t - \tau_{s,i}) \rightarrow PM_{2.5,i}(t)$$

This is a major improvement.

Now your model can learn:

"Fire activity increased in Punjab 8 hours ago, the atmospheric conditions transported the plume toward Delhi, and the corresponding Delhi stations should experience an increase now."

That is much closer to actual atmospheric physics.

Your research currently proposes
external source $\rightarrow$ Delhi station when wind blows NW $\rightarrow$ Delhi.

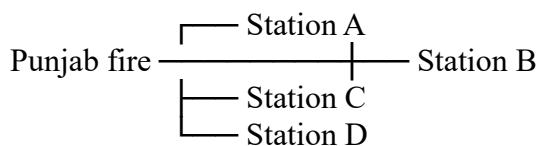
I would replace this with a

Probabilistic Transport Corridor

Instead of:

Punjab $\longrightarrow$ Delhi

use:



The probability of transport becomes:

$$P(i, | st) = f(w, i, n, d, d, i, stancedirectionBLHstabilityterrainhumidity)$$

This gives every source-receptor pair a **transport probability**.

That is much more sophisticated than a binary edge.

Add a "source contribution head"

This is probably the **single most important addition**

Don't only predict:

$$PM_{2.5} = 250 \mu\text{g}/\text{m}^3$$

Also predict:

Why is $PM_{2.5}$ 250?

For example

Source	Estimated contribution
Local traffic	31%
Delhi/NCR industry	18%
Punjab/Haryana biomass burning	29%
Dust	11%

Source	Estimated contribution
Regional background	11%

This changes your research from:
"another PM_{2.5} prediction model"
to:

Forecasting + source attribution + transport interpretation

That is substantially more valuable scientifically.

Delhi already has research systems attempting region- and sector-wise PM_{2.5} contribution analysis, including transport, NCR districts and stubble burning.

Your contribution should therefore **not claim source apportionment itself is new**. Instead, your novelty should be a **real-time source-to-receptor contribution forecasting integrated directly into the dynamic transport graph**.

Add a physics-conservation constraint

This is the upgrade I would strongly recommend for the Novelty. Instead of allowing the neural network to freely learn everything, impose a simplified atmospheric transport constraint.

Conceptually:

$$\frac{\partial C}{\partial t} = -u \cdot \nabla C + \nabla \cdot (K \nabla C) + E - R$$

where:

- C = PM_{2.5} concentration
- u = wind field
- K = effective diffusion
- E = emissions
- R = removal

You don't necessarily need to solve the full atmospheric chemistry equation.

Instead, use it as a **physics regularization loss**:

$$L = L_{forecast} + \lambda_1 L_{transport} + \lambda_2 L_{mass} + \lambda_3 L_{stability}$$

This makes the system **physics-constrained rather than purely data-driven**.

This direction is strongly supported by recent research: physics-informed GNNs have been shown to improve air-pollution forecasting, and transport-informed graph models are now explicitly learning advection/diffusion/reaction processes.

<https://proceedings.mlr.press/v277/assiotis25a.html>

Very important: your model should learn local + regional pollution simultaneously

Your research correctly recognizes the problem
reality = Region → Station
rather than only:
Station → Station.

I would formalize this into **three graph layers**.

Graph 1 — Local graph

Delhi monitoring stations:

$$G_L$$

captures:

- traffic
- local industry
- neighborhood accumulation
- local meteorology

Graph 2 — Regional transport graph

Punjab/Haryana/Rajasthan/NCR:

$$G_R$$

captures:

- biomass burning
- dust
- regional industrial emissions
- regional transport

Graph 3 — Atmospheric-state graph

A latent atmospheric state:

$$G_A$$

captures:

- BLH
- inversion
- humidity
- wind
- stability
- precipitation

Then:

$$G(t) = G_L(t) + G_R(t) + G_A(t)$$

This **multi-scale graph** is much stronger than a single station graph.

Add event-aware intelligence

Your document mentions sudden events such as crop burning and Diwali.
Make these into an explicit

Pollution Event Detector

The model first estimates:

$$P(Event_t)$$

for:

- biomass burning episode

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- dust episode
- fireworks episode
- inversion episode
- regional transport episode
- local accumulation episode

Then the forecasting network switches its operating mode.

For example:

Normal mode

Local + regional transport

Inversion mode

Local accumulation ↑

Vertical dispersion ↓

Biomass-burning mode

External source influence ↑

Transport corridor activated

Dust mode

Rajasthan/NW source influence ↑

PM10 influence ↑

This is much more interesting than simply adding a "festival indicator."

Your data strategy should also be upgraded

Your research already proposes CPCB, ERA5 and NASA FIRMS.

Keep these.

Core datasets

1. CPCB/CAAQMS

Hourly:

- PM_{2.5}
- PM₁₀
- NO₂
- CO
- O₃
- temperature
- RH
- wind

CPCB's public systems provide station-level air-quality data, and the government open-data platform provides hourly real-time AQI data.

2. ERA5

Use:

- BLH
- 10-m wind
- temperature
- dew point
- surface pressure
- precipitation
- radiation

ERA5 explicitly provides boundary-layer height.

3. NASA FIRMS

Use:

- VIIRS fire locations

- FRP
- confidence
- spatial clustering
- temporal evolution

FIRMS provides near-real-time global active-fire information from MODIS/VIIIRS.

I would add

4. Satellite aerosol information

Potentially:

- AOD
- aerosol layer information

5. Backward/forward trajectory information

For example:

- HYSPLIT-derived transport pathways

6. Land-use / road-network information

For:

- traffic
- industrial regions
- construction
- urban morphology

Don't treat FIRMS fire count as emission

Your document says:

Fire count from NASA FIRMS → crop-burning index.

That's okay as a predictor, but **do not call fire count "emission."**

FIRMS provides detected fire/thermal anomalies, not direct PM_{2.5} emission flux. NASA itself notes that satellite fire detections have spatial-resolution and detection limitations.

Instead derive:

Fire Emission Proxy

$$FEP = \sum_i FRP_i \times confidence_i \times area_i \times temporal\ weight$$

and we can call it something like

Satellite-derived biomass-burning activity index

rather than "actual emission."

That distinction will matter to reviewers.

Your existing BLH concept is scientifically justified

This part of your research is strong.

A Delhi study found that boundary-layer height is a major factor controlling aerosol accumulation and that low BLH during strong inversion conditions is associated with accumulation near the surface.

So your BLH concept is not arbitrary.

But the novelty should be:

How BLH and atmospheric stability dynamically control the learned source-to-receptor transport operator.

Not "We use BLH."

That second claim is too weak.

Don't overload the paper with

GAT + GRU + Transformer + CNN + LSTM + attention + reinforcement learning + XGBoost + random forest...

That will make the research look like an **AI component collection** rather than a scientific contribution.

Your architecture should remain conceptually clean

Transport physics → dynamic graph → neural operator → forecast

For example:

Dynamic Transport Encoder

↓

Physics-Constrained Graph Neural Operator

↓

Temporal Transformer/TCN

↓

Multi-task output

Your current document proposes:

Dynamic adjacency → GAT → GRU/TCN → PM_{2.5} forecast.

For the upgraded work, I recommend

Dynamic Transport Graph Neural Operator + Temporal Transformer

Not a conventional GAT.

Why?

Because your problem is fundamentally

transport and evolution of a pollutant field

rather than merely:

attention between neighboring stations.

The recent TransNet work is especially relevant because it demonstrates the value of explicitly representing advection, diffusion and reaction in a graph-based PM_{2.5} model.

<https://www.nature.com/articles/s44407-026-00052-x>

Therefore your research needs to **go beyond that direction**, not duplicate it.

The new model I recommend

STAGE-PM architecture

S — Source encoder

FIRMS + industrial + traffic + dust + event data

↓

T — Transport encoder

Wind + trajectory + distance + travel time

↓

A — Atmospheric stability encoder

BLH + inversion + RH + radiation + temperature

↓

G — Dynamic source-to-receptor graph

↓

E — Explainable physics-constrained neural operator

↓

Outputs

1. PM_{2.5} 1-hour
2. PM_{2.5} 6-hour
3. PM_{2.5} 24-hour
4. exceedance probability
5. source contribution
6. uncertainty
7. transport pathway explanation

That is the research direction I would suggest.

This is extremely important.

Do **not** simply compare your model against LSTM, GRU and GAT.

Your research needs **ablation experiments that prove each scientific hypothesis.**

Experiment A — Baseline

Historical PM_{2.5} + meteorology

Experiment B

- wind graph

Experiment C

- BLH

Experiment D

- inversion

Experiment E

- external source nodes

Experiment F

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- transport lag

Experiment G

- physics constraint

Experiment H

Full proposed model

This lets you demonstrate:

Which component actually produces the improvement?

Test specifically on extreme episodes

This is critical.

Your average MAE may look excellent while the model completely fails during severe pollution.

Therefore create:

Normal-condition dataset

$PM_{2.5} < \text{threshold}$

Moderate pollution

Severe pollution

Extreme episode

And separately:

- inversion episodes
- stubble-burning episodes
- dust episodes
- Diwali/fireworks
- regional transport episodes

Your paper should demonstrate:

The proposed model gains the largest advantage precisely when conventional models fail.

That is a much stronger scientific story.

Use spatial and temporal holdout

A very important reviewer concern will be leakage.

Do not randomly split hourly observations.

Use:

Temporal split

2018–2021 → training

2022 → validation

2023 → test

or an equivalent multi-year design.

And additionally:

Spatial generalization

Hold out selected monitoring stations.

Then ask:

Can the model forecast at a station it has never seen during training?

That would make the work considerably stronger.

Add uncertainty

Instead of:

$PM_{2.5} = 185$

produce:

$185 \pm \text{uncertainty interval}$

For operational air-quality forecasting, this is valuable.

You can output:

$$P(PM_{2.5} > 200)$$

rather than only a point forecast.

This is particularly useful for government decision support.

Your final research contribution should be framed differently

Weak framing

"We developed a GNN for $PM_{2.5}$ forecasting in Delhi."

Strong framing

"We develop a physics-constrained, source-aware dynamic transport graph that explicitly represents the evolving atmospheric pathways linking regional emission sources to urban receptors under changing atmospheric stability."

The scientific hypothesis

I recommend formally defining **three hypotheses**.

H1 — Stability hypothesis

Incorporating atmospheric stability into dynamic graph construction significantly improves $PM_{2.5}$ forecasting during shallow-boundary-layer episodes.

H2 — Transport hypothesis

Explicit source-to-receptor transport pathways and dynamic transport lag improve prediction during regional pollution episodes.

H3 — Physics hypothesis

Physics-constrained graph propagation improves generalization and extreme-event forecasting compared with purely data-driven STGNNs.

These are testable.

That's much better than simply saying "our model is accurate."

Do not start coding the current model yet.

First redesign the research around this core architecture:

Source → Transport → Stability → Receptor → Forecast → Attribution
and mathematically encode the transport process.

That will give you **three layers of contribution simultaneously**

Scientific contribution

A new physics-aware atmospheric transport learning framework.

Engineering contribution

An operational $PM_{2.5}$ forecasting system for Delhi-NCR.

IP contribution

A potentially patentable mechanism for **stability-adaptive source-to-receptor graph construction and transport-aware forecasting**.

This is considerably stronger than the current "**BLH-aware STGNN**" concept.